DTP AFRICA – COMPLETE SCOUTING INTELLIGENCE PLATFORM DOCUMENTATION

PRODUCT OVERVIEW

DTP Africa is a football scouting intelligence platform focused on discovering, profiling, analyzing, and matching grassroots football players to tactical systems, playing philosophies, and club requirements.

The system combines:

- Wearable/player tracking metrics
- Tactical intelligence scoring
- Philosophy fit analysis
- AI-driven Strong XI generation
- Position intelligence
- Player development analytics
- Club-fit scouting workflows
- Tactical role prediction
- Advanced scouting dashboards

This document defines the architecture, modules, scoring systems, database structure, APIs, UI requirements, and tactical engines required to integrate the Scouting Intelligence Layer into the existing DTP Africa platform.

CORE PRODUCT GOALS

The platform must allow:

1. Scouts to identify players fitting specific tactical systems.
2. Clubs to search for system-ready talent.
3. Automatic Strong XI generation by philosophy.
4. Tactical comparison between players.
5. Tactical role detection.
6. Development gap analysis.
7. Position suitability analysis.
8. Data-driven player recommendations.
9. Visual scouting reports.
10. AI-enhanced player profiling.

SYSTEM MODULES

The scouting intelligence platform contains the following modules:

1. Player Metrics Engine
2. Tactical Style Engine
3. Philosophy Fit Engine
4. Strong XI Generator
5. Tactical Match Engine
6. Position Suitability Engine
7. Scout Dashboard
8. Tactical Pitch Visualization Engine
9. Export & Reporting Engine
10. Player Comparison Engine
11. API Layer
12. Admin Management System
13. Benchmark Calibration System
14. Analytics & Aggregation Layer

TECH STACK RECOMMENDATION

Frontend

Recommended:

- Next.js
- React
- TailwindCSS
- Framer Motion
- Recharts
- Zustand or Redux
- React Query / TanStack Query

Backend

Recommended:

- Node.js
- NestJS or Express
- PostgreSQL
- Prisma ORM
- Redis
- BullMQ

Data Processing

Recommended:

- Python microservices for scoring engines
- Pandas
- NumPy
- FastAPI

Infrastructure

- Docker
- AWS / GCP
- CloudFront CDN
- S3 bucket storage
- PostgreSQL managed instance

DATABASE ARCHITECTURE

TABLE: players

Core player profile.

Fields:

id first_name last_name display_name date_of_birth age age_group nationality height_cm weight_kg dominant_foot position_primary position_secondary player_photo_url video_url biography created_at updated_at

---

TABLE: player_match_metrics

Stores match-level tracking metrics.

Fields:

id player_id match_id minutes_played ball_touches one_touch_possessions multi_touch_possessions possession_duration_seconds release_laces_count release_inside_foot_count release_other_footzone_count left_foot_touch_count right_foot_touch_count left_foot_release_count right_foot_release_count left_foot_receive_count right_foot_receive_count left_foot_kicking_power_mph right_foot_kicking_power_mph total_distance_miles sprint_distance_yards first_step_accelerations intense_accelerations kicking_power_mph top_speed_mph sprints intense_turns

avg_turn_entry_speed_mph avg_turn_exit_speed_mph left_turns right_turns back_turns distance_with_ball_yards top_speed_with_ball_mph intense_turns_with_ball created_at updated_at

---

TABLE: player_aggregated_metrics

Stores season-level or rolling averages.

Fields:

id player_id season matches_played minutes_played all metric averages created_at updated_at

---

TABLE: tactical_scores

Stores calculated tactical scores.

Fields:

id player_id gegenpress_score tiki_taka_score counter_attack_score total_football_score high_line_score winger_score low_block_score quick_release_score ball_retention_score technical_cleanliness_score direct_play_score flair_score press_resistance_score possession_hub_score transition_explosiveness_score isolation_threat_score two_footedness_score created_at updated_at

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TABLE: philosophy_fit_profiles

Stores philosophy-specific fit scores.

Fields:

id player_id formation philosophy position_slot philosophy_fit_score position_fit_score confidence_score created_at updated_at

---

TABLE: tactical_benchmarks

Professional benchmark profiles.

Fields:

id player_name position country club_style tactical_archetype dominant_foot gegenpress_score tiki_taka_score counter_attack_score total_football_score high_line_score winger_score low_block_score best_system_fit description is_active created_at updated_at

METRICS ENGINE

PURPOSE

Convert raw tracking data into:

- Tactical scores
- Technical scores
- Intelligence metrics
- Position suitability metrics
- Tactical fit scores

DERIVED METRICS

turn_efficiency

Formula:

turn_efficiency = avg_turn_exit_speed_mph / avg_turn_entry_speed_mph

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turn_diversity

Formula:

turn_diversity = (left_turns + right_turns + back_turns) / 3

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workload_index

Formula:

workload_index = (total_distance_miles * 1609 + sprint_distance_yards * 0.9144) / minutes_played

---

sprint_frequency

Formula:

sprint_frequency = sprints / minutes_played

acceleration_density

Formula:

acceleration_density = (first_step_accelerations + intense_accelerations) / minutes_played

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ball_activity_rate

Formula:

ball_activity_rate = ball_touches / minutes_played

---

two_footedness_score

Formula:

two_footedness_score = 100 - ABS(left_foot_touch_pct - right_foot_touch_pct)

---

quick_release_score

Formula:

quick_release_score = (one_touch_possessions / total_possessions) * 100

---

technical_cleanliness_score

Formula:

technical_cleanliness_score = (release_inside_foot_count + controlled_receive_actions + turn_efficiency)

---

direct_play_score

Formula:

direct_play_score = (release_laces_count + kicking_power_mph + sprint_distance_yards)

---

flair_score

Formula:

flair_score = (release_other_footzone_count + intense_turns_with_ball + turn_diversity)

TACTICAL STYLE ENGINE

PURPOSE

Calculate tactical philosophy compatibility.

GEGENPRESSING SCORE

Formula:

gegenpress_score = (intense_accelerations * 0.30 + sprints * 0.25 + intense_turns * 0.20 + sprint_distance_yards * 0.15 + workload_index * 0.10)

TIKI-TAKA SCORE

Formula:

tiki_taka_score = (ball_touches * 0.35 + turn_efficiency * 0.25 + distance_with_ball_yards * 0.20 + (left_turns + right_turns) * 0.20)

COUNTER ATTACK SCORE

Formula:

counter_attack_score = (top_speed_mph * 0.30 + sprint_distance_yards * 0.25 + top_speed_with_ball_mph * 0.20 + distance_with_ball_yards * 0.15 + kicking_power_mph * 0.10)

TOTAL FOOTBALL SCORE

Formula:

total_football_score = (total_distance_miles * 0.30 + turn_diversity * 0.25 + distance_with_ball_yards * 0.20 + sprint_frequency * 0.15 + acceleration_density * 0.10)

HIGH LINE SCORE

Formula:

high_line_score = (top_speed_mph * 0.40 + sprint_frequency * 0.30 + back_turns * 0.20 + intense_accelerations * 0.10)

WINGER SCORE

Formula:

winger_score = (top_speed_with_ball_mph * 0.30 + intense_turns_with_ball * 0.25 + distance_with_ball_yards * 0.25 + sprints * 0.20)

LOW BLOCK SCORE

Formula:

low_block_score = (intense_turns * 0.35 + first_step_accelerations * 0.25 + back_turns * 0.20 + (1 / total_distance_miles) * 0.20)

PHILOSOPHY FIT ENGINE

PURPOSE

Allow scouts to select:

- Playing philosophy
- Formation

Then automatically generate:

- Strong XI
- Best-fit players
- Tactical fit rankings

SUPPORTED PHILOSOPHIES

1. Gegenpressing
2. Tiki-Taka
3. Counter-Attacking
4. Total Football
5. High Line System
6. Low Block System
7. Wing Play
8. Direct Vertical Football
9. Inverted Fullback System
10. False 9 Fluid System

STRONG XI GENERATOR

PURPOSE

Automatically generate:

- Best tactical XI
- Bench alternatives
- Tactical summaries

PROCESS

1. Select formation.
2. Select philosophy.
3. Load eligible players.
4. Calculate philosophy fit score.
5. Apply positional weighting.
6. Rank players.
7. Select strongest player per role.
8. Generate bench alternatives.
9. Generate tactical insights.

PHILOSOPHY FIT SCORE

Formula:

philosophy_fit_score = (primary_style_score * 0.50 + overall_rating * 0.25 + position_fit_score * 0.15 + offensive_or_defensive_relevance * 0.10)

FORMATIONS

Supported:

4-3-3 4-2-3-1 4-4-2 3-4-3 3-5-2 4-1-4-1 5-3-2

SOCCER PITCH VISUALIZATION ENGINE

PURPOSE

Render:

- Tactical formations
- Strong XI
- Interactive player markers

PITCH UI REQUIREMENTS

- Green pitch background
- White field markings
- Responsive scaling
- Animated player markers
- Tactical formation spacing
- Hover/click interactions

PLAYER MARKER CONTENT

Each player marker must show:

- Player photo
- Player name
- Position
- Fit score
- Overall rating
- Offensive rating

- Defensive rating

PLAYER DETAIL DRAWER

On click, open side drawer.

Display:

- Full profile
- Tactical scores
- Physical metrics
- Technical metrics
- Tactical suitability
- Video links
- Scout notes
- Tactical tags

PLAYER ARCHETYPE TAGS

Generate automatically.

Examples:

- One-Touch Specialist
- Press Resistant Midfielder
- Vertical Transition Threat
- Elite Two-Footed
- Possession Hub
- Isolation Specialist

TACTICAL MATCH ENGINE

PURPOSE

Compare players against benchmark tactical profiles.

Return:

- Nearest tactical matches
- Similarity scores

- Tactical archetypes
- Tactical gaps

SIMILARITY ENGINE

Use weighted Euclidean similarity.

SCOUT DASHBOARD

FEATURES

- Tactical search
- Philosophy filtering
- Position filters
- Tactical score filters
- Strong XI generation
- Player comparison
- Tactical reports

FILTERS

Support:

- Position
- Age group
- Tactical fit score
- Offensive rating
- Defensive rating
- Dominant foot
- Tactical archetype

PLAYER COMPARISON ENGINE

PURPOSE

Compare up to 4 players.

COMPARISON METRICS

Compare:

- Tactical scores
- Physical metrics
- Technical metrics
- Tactical fit scores
- Offensive metrics
- Defensive metrics

VISUALIZATIONS

Use:

- Radar charts
- Bar charts
- Scatter plots
- Percentile displays

POSITION SUITABILITY ENGINE

PURPOSE

Recommend ideal positions.

POSITIONAL FIT FACTORS

Use:

- Tactical scores
- Speed profile
- Technical profile
- Press resistance
- Direct play profile
- Flair profile
- Work rate

EXAMPLE OUTPUT

{ "primary_position_fit": "RW", "secondary_position_fit": "LW", "hybrid_fit": "AM", "confidence": 88 }

EXPORT ENGINE

SUPPORTED EXPORTS

- PDF
- CSV
- Shareable links

PDF EXPORT CONTENT

Include:

- Strong XI pitch image
- Tactical summary
- Player cards
- Tactical fit scores
- Tactical insights
- Bench alternatives

API ARCHITECTURE

BASE ROUTE

/api

PLAYER ENDPOINTS

GET /api/players GET /api/players/:id GET /api/players/:id/tactical-scores GET /api/players/:id/position-fit GET /api/players/:id/tactical-profile

---

SCOUTING ENDPOINTS

GET /api/scouting/philosophy-fit GET /api/scouting/strong-xi GET /api/scouting/player-comparison GET /api/scouting/tactical-search

---

BENCHMARK ENDPOINTS

GET /api/benchmarks POST /api/benchmarks PUT /api/benchmarks/:id DELETE /api/benchmarks/:id

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ADMIN SYSTEM

PURPOSE

Allow admins to:

- Manage benchmarks
- Recalculate scores
- Edit player profiles
- Trigger recalibration jobs
- Manage tactical mappings

BACKGROUND JOBS

Use queue workers for:

- Tactical recalculations
- Aggregation updates
- Benchmark updates
- Report generation

PERFORMANCE OPTIMIZATION

REQUIREMENTS

- Cache tactical scores
- Cache Strong XI queries
- Paginate search results

- Precompute tactical fit scores
- Use indexed database fields

RECOMMENDED INDEXES

Create indexes on:

player_id position_primary age_group gegenpress_score tiki_taka_score counter_attack_score total_football_score winger_score high_line_score low_block_score overall_rating

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SECURITY

REQUIREMENTS

- JWT authentication
- Role-based permissions
- Scout access controls
- Admin-only recalculation endpoints
- Rate limiting

USER ROLES

1. Admin
2. Scout
3. Club Representative
4. Analyst
5. Read-only Viewer

ANALYTICS LAYER

DASHBOARD METRICS

Track:

- Most searched players
- Most used philosophy
- Top tactical archetypes
- Tactical score distributions

- Regional talent trends

UI DESIGN SYSTEM

BRAND STYLE

Use:

- Dark modern dashboard UI
- Green tactical pitch
- Rounded cards
- Clean typography
- Tactical heatmap colors
- High-contrast data visualization

COLOR SYSTEM

Primary:

2563eb

Dark background:

1e293b

Pitch green:

166534

Elite score:

22c55e

Warning:

f59e0b

Danger:

ef4444

MOBILE RESPONSIVENESS

The system must:

- Work on tablet
- Work on mobile
- Maintain pitch usability
- Collapse filters responsively
- Support touch interactions

FUTURE EXPANSION

Planned future modules:

1. Injury risk prediction
2. AI match simulation
3. Club tactical compatibility engine
4. Recruitment recommendation engine
5. Video event tagging AI
6. GPS live tracking ingestion
7. Match event intelligence
8. xG/xA integration
9. Team chemistry engine
10. Tactical chemistry simulation

DEVELOPMENT PRIORITY ORDER

PHASE 1

- Database expansion
- Metrics ingestion
- Tactical score engine
- Tactical APIs

PHASE 2

- Philosophy Fit Scout page

- Pitch visualization
- Strong XI generator

PHASE 3

- Tactical comparison engine
- Export engine
- Admin system

PHASE 4

- AI tactical recommendations
- Benchmark calibration
- Advanced analytics

FINAL PRODUCT OBJECTIVE

The final system should function as:

A full football tactical intelligence platform capable of:

- Understanding player styles
- Matching players to tactical systems
- Helping clubs recruit smarter
- Helping scouts identify fit faster
- Helping players understand development gaps
- Creating AI-driven football intelligence workflows

The platform should feel like a modern fusion of:

- Wyscout
- Hudl
- StatsBomb
- Playermaker
- FM Scouting
- Tactical analyst software

built specifically for African grassroots football intelligence.